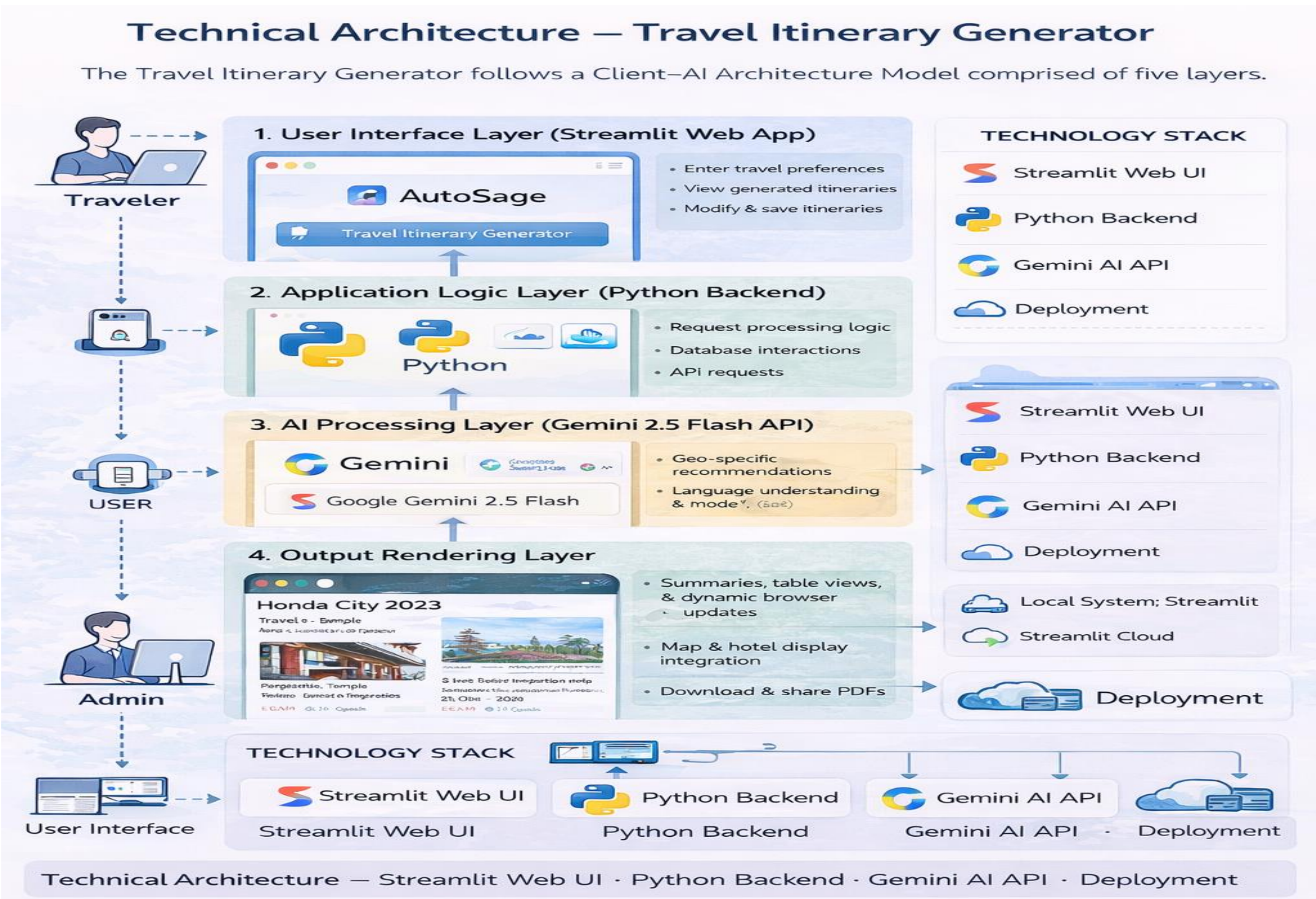
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 Feb 2026
Team ID	LTVIP2026TMIDS90611
Project Name	Travel-Itinerary-Generator
Maximum Marks	4 Marks

Technical Architecture:

Travel-Itinerary-Generator follows a Client–AI Architecture Model consisting of:

1. User Interface Layer (Streamlit Web App)
2. Application Logic Layer (Python Backend)
3. AI Processing Layer (Gemini 2.5 Flash API)
4. Output Rendering Layer
5. Deployment Layer (Local / Streamlit Cloud)



Technical Architecture — Travel Itinerary Generator
The Travel Itinerary Generator follows a Client–AI Architecture Model comprised of five layers.
Traveler
1. User Interface Layer (Streamlit Web App)
AutoSage
Travel Itinerary Generator
• Enter travel preferences
• View generated itineraries
• Modify & save itineraries
TECHNOLOGY STACK
Streamlit Web UI
Python Backend
Gemini AI API
Deployment
2. Application Logic Layer (Python Backend)
Python
• Request processing logic
• Database interactions
• APi requests
3. AI Processing Layer (Gemini 2.5 Flash API)
Gemini
Google Gemini 2.5 Flash
• Geo-specific recommendations
• Language understanding & mode
USER
Streamlit Web UI
Python Backend
Gemini AI API
Deployment
4. Output Rendering Layer
Honda City 2023
• Summaries, table views, & dynamic browser updates
• Map & hotel display integration
• Download & share PDFs
Local System: Streamlit
Streamlit Cloud
Deployment
Admin
TECHNOLOGY STACK
Streamlit Web UI
Python Backend
Gemini AI API
User Interface
Streamlit Web UI
Python Backend
Gemini AI API
Deployment
Technical Architecture — Streamlit Web UI · Python Backend · Gemini AI API · Deployment

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based interface for uploading images and interacting with assistant	Streamlit (Python-based Web Framework)
2.	Application Logic-1	Image processing logic and prompt construction	Python
3.	Application Logic-2	Conversational query processing and structured prompt generation	Python
4.	Application Logic-3	Session state handling and UI mode switching (Image / Assistant)	Streamlit Session State
5.	File Storage	Temporary image storage during session	In-memory processing (PIL / Local Runtime Memory)
6.	External API-1	Multimodal AI processing for image & text analysis	Google Gemini 2.5 Flash API
7.	Machine Learning Model	Multimodal Generative AI for vehicle identification and advisory	Gemini 2.5 Flash
8.	Infrastructure (Server / Cloud)	Local development and optional cloud deployment	Local System / Streamlit Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Application built using open-source libraries and frameworks	Python, Streamlit, Pillow, python-dotenv
2.	Security Implementations	Secure API key handling using environment variables; no hardcoded credentials	.env file, Streamlit Secrets, OS Environment Variables
3.	Scalable Architecture	Modular architecture separating UI, logic, and AI layers; supports feature expansion	API-Based AI Integration (Gemini)
4.	Availability	Web-based deployment accessible via browser; depends on cloud hosting & API availability	Streamlit Cloud / Local Server
5.	Performance	Optimized prompt engineering and lightweight UI; average 7–12 sec response time	Gemini Flash (Fast Inference Model)